

Infosys Springboard Virtual Internship 7.0 Completion Report

Team Details

Batch Number : Team-A

Start date : 21 July 2026

Names: 1.Azmeera Raghavendar

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3.Ankitha KS

4.Esha Yadav

Internship Duration: 8 Weeks

1. Project Title

AI-Based Industrial Maintenance and Process Optimization System

2. Project Objective

The main objective of the **AI-Based Industrial Maintenance and Process Optimization System** is to develop an intelligent platform that monitors industrial machines, analyzes operational data, and predicts potential machine failures in advance.

The system aims to:

- Monitor machine health using real-time and historical sensor data.
- Analyze parameters such as temperature, vibration, pressure, RPM, humidity, voltage/current, oil level, and load.
- Use machine learning techniques to identify abnormal machine conditions and predict potential failures.
- Classify machine conditions into Normal, Possible Failure, and High Risk.
- Provide a centralized dashboard for machine monitoring, telemetry, maintenance, incidents, inventory, and notifications.
- Support proactive maintenance and process optimization to reduce unexpected downtime and improve machine reliability.
- Help maintenance teams make data-driven decisions and improve overall industrial operational efficiency.

3. Project description in detail

The AI-Based Industrial Maintenance and Process Optimization System is an intelligent industrial monitoring platform developed to improve machine maintenance, operational efficiency, and process management. The system combines machine monitoring, sensor telemetry, maintenance management, incident tracking, inventory management, notifications, and AI-based prediction in a centralized platform.

The system collects and analyzes important machine parameters such as temperature, vibration, pressure, RPM, humidity, voltage/current, oil level, and load. This information is presented through a dashboard, allowing users to monitor machine conditions and identify abnormal operating behavior. The system uses a Random Forest machine learning model to analyze machine condition data and predict potential failures. Based on the prediction results, machines are classified into Normal, Possible Failure, and High Risk categories. This helps maintenance teams identify machines that may require inspection or preventive maintenance.

The application is developed using Streamlit and Python for the frontend and FastAPI for backend REST APIs. SQLite with SQLAlchemy is used for database management, while Pandas, NumPy, and Scikit-learn are used for data processing and machine learning. Plotly is used for data visualization, and Swagger/OpenAPI supports API documentation and testing.

The system also manages maintenance schedules, incidents, inventory levels, and notifications, providing a complete view of industrial operations. By combining monitoring and predictive analytics, the project supports proactive maintenance, reduces the possibility of unexpected downtime, improves machine reliability, and assists in data-driven maintenance decisions. The project demonstrates how Artificial Intelligence, Machine Learning, data analytics, and full-stack application development can be integrated to address real-world industrial maintenance and process optimization requirements.

4. Timeline Overview

Week	Activities Planned	Activities Completed
Week 1	Understand the problem statement, project requirements, predictive maintenance concepts, and existing industrial maintenance systems.	Studied predictive maintenance and industrial machine monitoring. Finalized the project title, objectives, requirements, and overall system concept.
Week 2	Analyze system requirements and design the overall architecture, database structure, and application modules.	Designed the system architecture and identified the major modules: Dashboard, Robots, Sensors, Telemetry, Maintenance, Incidents, Inventory, Notifications, and Prediction.
Week 3	Develop the backend services and database for storing machine, telemetry, maintenance, and incident information.	Implemented the FastAPI backend and SQLite database. Created REST API endpoints and database models for machine monitoring and maintenance operations.
Week 4	Develop the frontend dashboard and machine monitoring interfaces.	Developed the Streamlit frontend with dashboard, robot/machine monitoring, sensor, telemetry, maintenance, and incident interfaces.
Week 5	Collect and prepare machine	Prepared telemetry data

	telemetry data and develop the machine learning prediction module.	containing parameters such as temperature, vibration, pressure, RPM, humidity, voltage/current, and load. Developed and trained machine learning models for risk prediction.
Week 6	Integrate the trained ML model with the application and implement risk classification and maintenance features.	Integrated the Random Forest model with the system. Implemented machine risk classification into Normal, Possible Failure, and High Risk and connected prediction results with the dashboard.
Week 7	Test the complete system, identify errors, and improve system performance and usability.	Tested frontend, backend APIs, database operations, prediction module, maintenance records, incidents, inventory, and notifications. Fixed identified issues and improved the user interface.
Week 8	Perform final validation, document the project, prepare screenshots and presentation, and demonstrate the completed system.	Completed system validation and documentation. Prepared project report, PPT, screenshots, results, future scope, and final project demonstration.

5a. Key Milestones

Milestone	Description	Date Achieved
Project Kickoff	Finalized the project topic, objectives, requirements, and initial plan for developing the AI-based industrial predictive maintenance system.	30 July 2026
Prototype/First Draft	Developed the initial prototype with the basic frontend, backend, database, machine monitoring, and telemetry modules.	10 August 2026
Mid-Term Review	Demonstrated the developed modules, reviewed system progress, tested the initial prediction model, and incorporated feedback.	21 August 2026
Final Submission	Completed the integrated system including machine monitoring, telemetry analysis, maintenance management, incident tracking, inventory, notifications, and AI-based risk prediction.	4 September 2026
Presentation	Prepared and presented the final project demonstration, results, system architecture, future scope, and conclusion.	16 September 2026

5b. Project execution details

The project execution began with understanding the requirements of an industrial predictive maintenance system and identifying the major machine parameters required for monitoring. The system was planned to monitor parameters such as temperature, vibration, pressure, RPM, humidity, voltage/current, and load. Based on these requirements, the overall system architecture and major modules were designed, including machine monitoring, telemetry, maintenance, incidents, inventory, notifications, and prediction.

The development phase focused on building the application components and database. A FastAPI-based backend was developed to provide REST APIs for managing machine and operational data, while SQLite was used for data storage. A Streamlit-based frontend was developed to provide an interactive dashboard for viewing machine health, sensor readings, telemetry information, maintenance schedules, incidents, inventory, and notifications.

The machine learning component was developed using machine telemetry data to identify abnormal machine conditions and predict potential failures. Different machine learning approaches were considered, and the Random Forest algorithm was used for risk prediction. The trained model was integrated into the application to classify machine conditions into Normal, Possible Failure, and High Risk categories.

The system integration phase involved connecting the frontend, backend, database, and machine learning components into a single working application. REST APIs were used for communication between the frontend and backend, allowing the dashboard to retrieve and display updated machine and operational information. The prediction module was also connected to the telemetry data so that machine risk levels could be generated based on operating conditions. Testing and validation were carried out to ensure that the developed modules worked correctly and reliably. The application was tested for machine monitoring, telemetry visualization, maintenance management, incident tracking, inventory management, notifications, and AI-based prediction. Identified errors and inconsistencies were resolved during testing, and the user interface was refined to improve usability and clarity.

Finally, the completed system was validated through overall functional testing and demonstration. Project documentation was prepared along with system screenshots, architecture details, results, future scope, and conclusion. The final application demonstrated how machine data and machine learning can be combined to support early identification of potential failures and improve industrial maintenance planning.

6. Snapshots / Screenshots

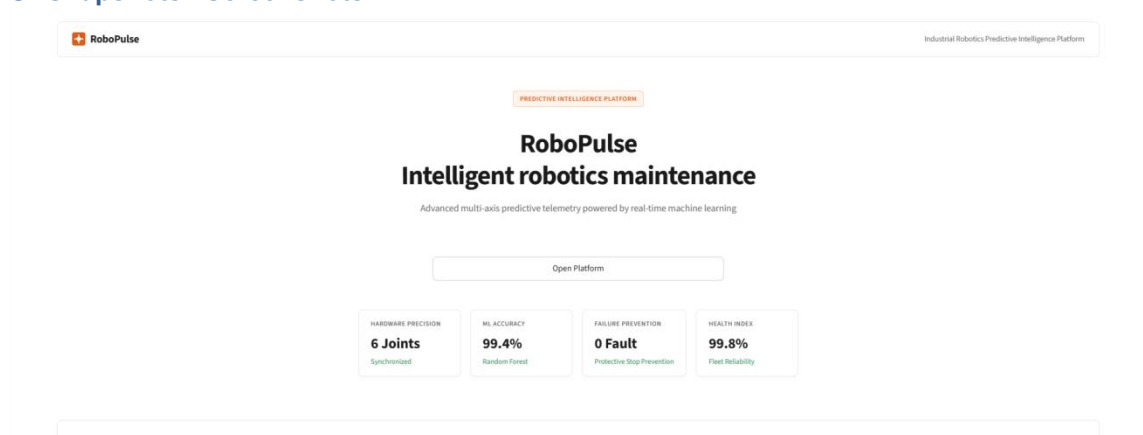


Fig1:Landing Page

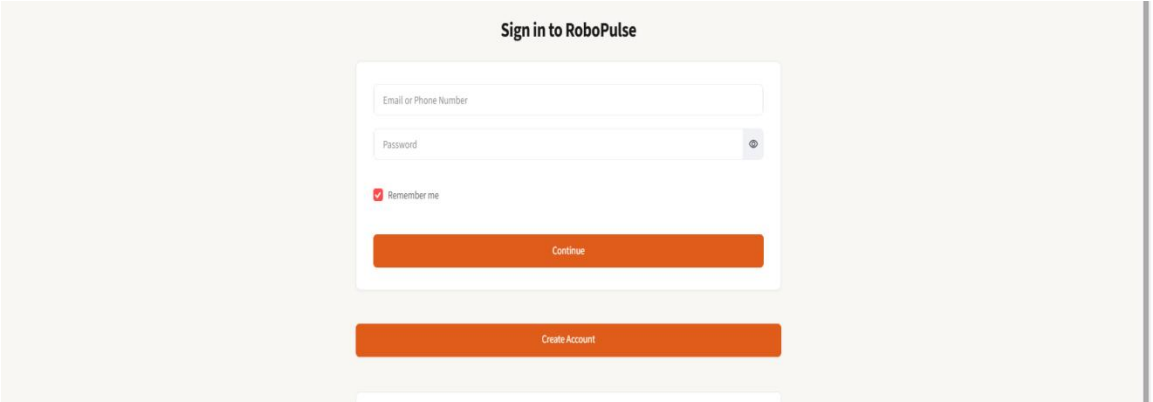


Fig2:Login/signup

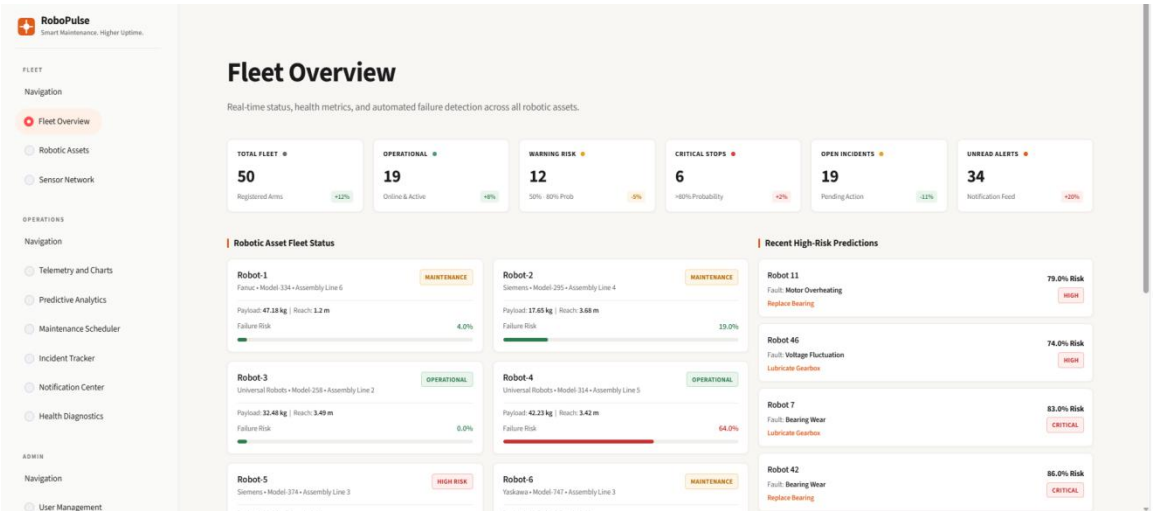


Fig3:Dashboard Overview

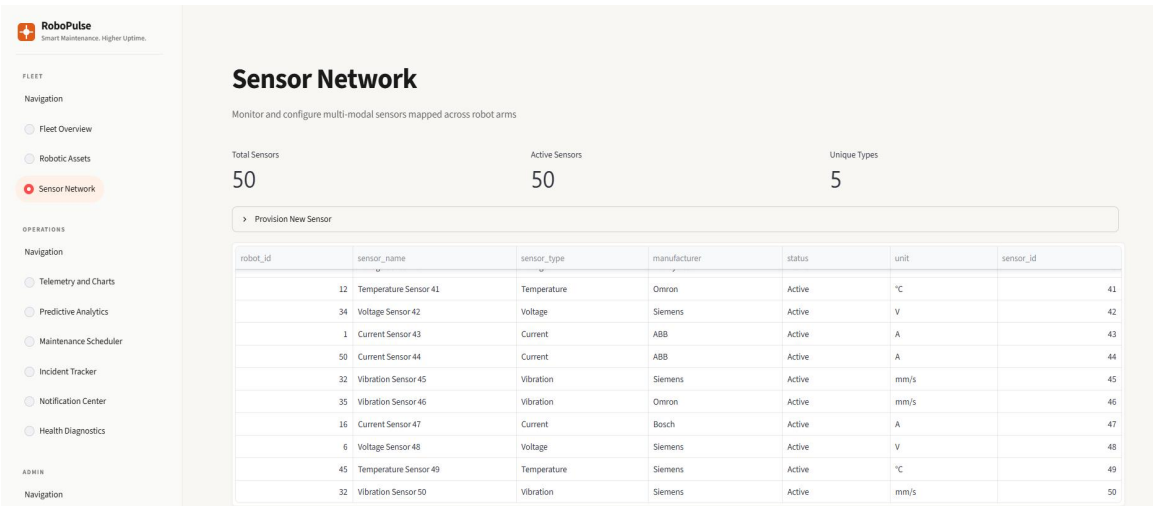


Fig4:Sensor Network

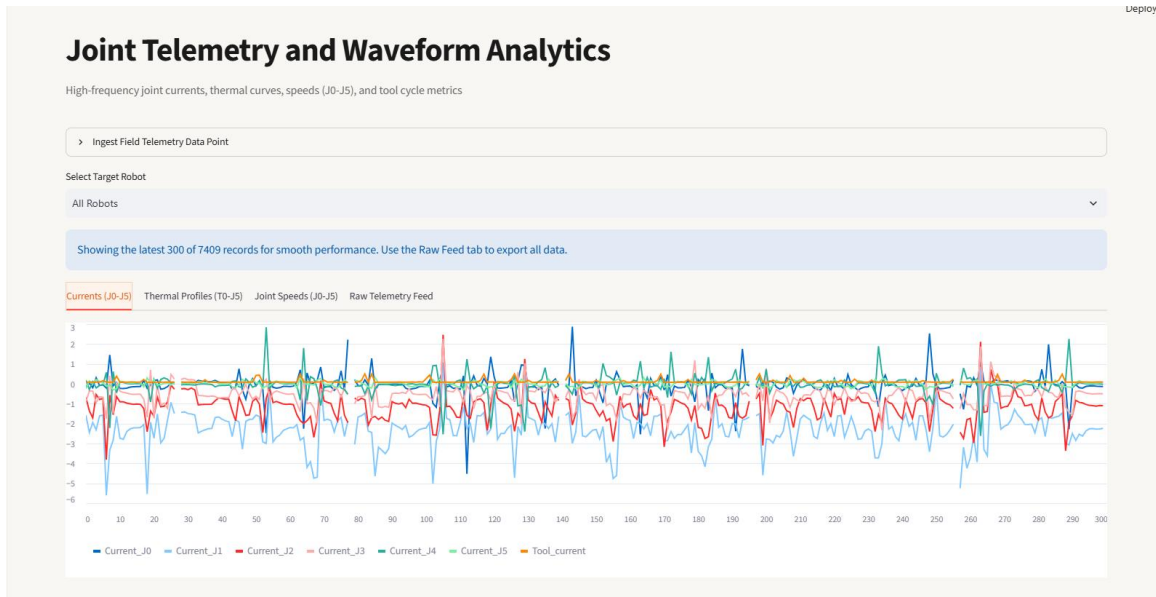


Fig5:Telemetry and Charts

Deploy

Predictive Analytics and Machine Learning

Automated protective stop forecasting, parameter simulation, and prescriptive recommendations

Automated Robot Diagnosis
Interactive ML Simulator
Historical Predictions Log

Interactive Protective-Stop Simulator

Adjust telemetry values to run the trained model against a simulated robot cycle.

Simulated Robot

Robot 1 - Robot-1

Adjust simulated telemetry

Current J0 -0.05 - +	Current J4 -0.06 - +	Speed J0 -0.05 - +
Temperature T0 37.19 - +	Temperature J4 45.31 - +	Speed J1 0.00 - +
Current J1 -2.85 - +	Current J5 0.05 - +	Speed J2 0.00 - +
Temperature J1 40.31 - +	Temperature J5 44.56 - +	Speed J3 0.01 - +
Current J2 -0.83 - +	Tool current 0.08 - +	Speed J4 0.00 - +

Fig6:AI/ML Prediction

7. Challenges Faced

- Preparing and handling machine telemetry data for accurate analysis and prediction.
- Selecting and training a suitable machine learning model for predicting machine failures.
- Integrating the machine learning model with the FastAPI backend and Streamlit frontend.
- Ensuring proper communication between the frontend, backend, database, and prediction modules.
- Managing and displaying machine sensor data effectively.
- Testing and debugging different modules after system integration.
- Maintaining consistent data across machine monitoring, maintenance, incidents, and inventory modules.
- Completing development, testing, documentation, and presentation within the given timeline.

8. Learnings & Skills Acquired

- Gained practical knowledge of Artificial Intelligence and Machine Learning concepts used in predictive maintenance.
- Learned to prepare, analyze, and process industrial machine telemetry and sensor data.
- Gained experience in developing machine learning models using Scikit-learn, including Random Forest.
- Learned to develop REST APIs using FastAPI and work with backend services.
- Gained practical experience in developing interactive dashboards using Streamlit, Python, HTML/CSS, and Plotly.
- Learned to manage and store application data using SQLite and SQLAlchemy.
- Gained knowledge of machine health monitoring, failure prediction, and maintenance management.
- Improved skills in API integration, database management, testing, debugging, and system integration.
- Developed better understanding of software development workflows and project documentation.
- Improved problem-solving, teamwork, communication, time management, and presentation skills.

9. Testimonials from team

- Successfully developed an AI-based predictive maintenance system for industrial machine monitoring.
- Demonstrated effective use of machine learning for identifying potential machine failures.
- Successfully integrated frontend, backend, database, and machine learning components into a unified system.
- Completed the assigned project activities within the internship timeline.
- Demonstrated good problem-solving, teamwork, communication, and technical skills throughout the

project.

- Gained practical experience in developing and testing a real-world software application.

10. Conclusion

The internship provided a valuable opportunity to apply theoretical knowledge to the development of a real-world software solution. The project, “AI-Based Industrial Maintenance and Process Optimization System,” focused on monitoring industrial machines, analyzing telemetry data, managing maintenance activities, and predicting potential machine failures using machine learning. Through the development process, I gained practical experience in Python, Streamlit, FastAPI, REST APIs, SQLite, SQLAlchemy, Scikit-learn, Pandas, NumPy, and Plotly.

The project helped me understand the complete software development process, starting from requirement analysis and system design to implementation, machine learning model development, integration, testing, debugging, and documentation. Working with machine parameters such as temperature, vibration, pressure, RPM, humidity, voltage/current, and load provided practical knowledge of how industrial data can be used for machine health monitoring and predictive maintenance.

The internship also improved my technical and professional skills. I developed better problem-solving abilities, learned to identify and resolve technical issues, and gained experience in integrating different technologies into a single application. The experience of working on different modules and completing the project within the given timeline also strengthened my teamwork, communication, time management, and presentation skills.

Overall, the internship helped bridge the gap between academic learning and practical industry-oriented development. It provided a better understanding of how Artificial Intelligence and Machine Learning can be applied to industrial applications to support proactive maintenance and improve operational efficiency. The knowledge and experience gained during this internship will be useful for my future academic projects and career goals in Artificial Intelligence, Machine Learning, Data Analytics, and Software Development.

11. Acknowledgements

I would like to express my sincere gratitude to Infosys Springboard for providing me with the opportunity to undertake this internship and gain valuable practical experience in Artificial Intelligence and Machine Learning. I am thankful to my mentor and trainers for their continuous guidance, support, valuable feedback, and encouragement throughout the internship.

I would also like to thank my project team members for their cooperation, coordination, and support during the development of the project. Their contributions and discussions helped in completing the project successfully.

I am grateful to my college faculty members for their encouragement and support throughout the internship. Finally, I would like to thank everyone who directly or indirectly contributed to my learning and helped me successfully complete this internship and project.